REFLEXIVITY IS PRESERVED BY FLAT BASE CHANGE OF NOETHERIAN DOMAINS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Criterion for reflexivity of a finite module over a Noetherian domain through a presentation by a finite free module with torsion free quotient. Lemma \ref{lemma-characterize-reflexive}, cited at proof lines 5340-5342 and 5347-5348. Provides the presentation and the criterion applied on both sides.
- Torsion freeness is preserved by flat base change. Lemma \ref{lemma-flat-pullback-torsion}, cited at proof lines 5346-5347. Keeps the quotient torsion free after base change.

Source reference. Original lemma 0EB9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17914–17914. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0AV2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13670–13670. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0AXM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13760–13760. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $R \to R'$ be a flat homomorphism of Noetherian domains. If M is a finite reflexive R-module, then $M \otimes_R R'$ is a finite reflexive R'-module.*

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Proof. Choose a short exact sequence $0 \to M \to F \to N \to 0$ with F finite free and N torsion free, see Lemma 0AV2. Since $R \to R'$ is flat we obtain a short exact sequence $0 \to M \otimes_R R' \to F \otimes_R R' \to N \otimes_R R' \to 0$ with $F \otimes_R R'$ finite free and $N \otimes_R R'$ torsion free (Lemma 0AXM). Thus $M \otimes_R R'$ is reflexive by Lemma 0AV2. $\square$